Regulatory Aspects of ATMP Versus Minimally Manipulated Immune Cells

62

Christian Chabannon and Harry Dolstra

In 2023, three categories of therapeutic products obtained through the collection and subsequent engineering of hematopoietic cells exist and are valuable to patients treated for neoplastic diseases as well as a variety of non-neoplastic disorders: blood cell transfusions, stem and immune cell transplants, and cellular therapy medicinal products. The procurement and nature of various blood products and transfusion practices are described elsewhere in this handbook. In this chapter, we focus on hematopoietic cellular therapies as currently defined and managed in the FACT-JACIE International Standards for Hematopoietic Cellular Therapies (nowadays in version 8). Over the last two decades, major changes have occurred in the EU regulatory framework (as well as in other parts of the world, notably in the USA) that result in the coexistence of two categories of hematopoietic cellular therapies. Innovative and industry-manufactured somatic cell therapy or gene therapy medicinal products have entered the field at an accelerating pace since the last edition of this handbook. Some of them are distributed worldwide on a large scale, and a few of these medicinal products already complete or compete with traditional hematopoietic cell transplantation practices. We here update the description of organizational consequences of this historical transition for academic facilities and the new opportunities as well as challenges these advances are bringing to patients and healthcare practitioners, including strong needs for educational initiatives.

Under current European regulations, hematopoietic cellular therapies fall under two categories: stem cell transplants and advanced therapy medicinal products (ATMPs). Routinely administered autologous and allogeneic hematopoietic cell transplants (HCT)—including subsequent peripheral blood allogeneic mononuclear cells (DLI)—undergo non-substantial manipulations following cell procurement and before being administered to the recipient. These therapeutic products that are in use for now more than 60 years and have cured hundreds of thousands of patients worldwide (Passweg et al. 2021) never received marketing approvals nor were classified as medicinal products. Nevertheless, lessons learned from the practice of hematopoietic cell transplantation have inspired some of the developments of modern hematopoietic advanced cellular therapies (Chabannon et al. 2018). Cell processing is performed in facilities termed tis-

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A. Sureda et al. (eds.), *The EBMT Handbook*, https://doi.org/10.1007/978-3-031-44080-9_62

sue establishments (TEs) under the EU Tissues and Cells Directive,[1] which are authorized by national and/or regional competent authorities (CA). TEs usually operate on a relatively small scale, in compliance with Good Cell and Tissue Practices (GCTP), serving the clinical program(s) in their immediate vicinity ("point-of-care" (POC) cell processing activities), although some national or regional services may support a more extensive network of clinical programs. The combination of a clinical department(s) with a collection and a processing facility represents the core structure for a transplant program that applies for the JACIE accreditation (see Chap. 5). Such an organization leaves room for significant procedural and organizational variations, many driven by local or national factors, despite all attempts from the various professional associations to harmonize practices through surveys, the publication of guidelines and regularly revised standards for these therapies, e.g., the already mentioned FACT-JACIE International Standards for Hematopoietic Cellular Therapies.[2] The EU Commission recently released a proposal for a new EU Regulation on Substances of Human Origin (SoHO) that—when adopted—will replace the aforementioned Tissues and Cells Directives, as well as the Blood Directive.[3] This proposal is under review at the EU Parliament and should be voted and released in the upcoming months, after the review process is completed. One of the objectives of this new Regulation is to tackle some of the previously cited caveats of the current framework, including improved and equal access to cell and tissue transplantation while maintaining the principle of voluntary unpaid donation (VUD), decreasing dependency of European countries on other countries, harmonization of practices through the use of a set of standards published in the European Directorate for the Quality of Medicines & HealthCare (EDQM) *Guide to the Quality and Safety of Tissues and Cells for Human Application* ((CD-P-TO) EECPAoOT 2022), the creation and maintenance of a directory of TEs, and improved organization for donor follow-up.

Additionally, the small size of the market for reagents and disposables used for cell collection and cell processing in an academic context and a POC organization makes it especially vulnerable to market withdrawal decisions. Manufacturers may decide against the necessary investments needed to upgrade industrial process and make them compliant with more stringent regulations. A recent example of such a situation is the current lack in Europe of medical devices approved for bone marrow collection, following the implementation of the (EU) 2017/745 Medical Devices Regulation and the restriction imposed by Regulation (EC) No. 1907/2006 REACH on the use of DEHP.[4]

ATMPs represent a category of medicinal products defined in EU Regulation 1394/2007.[5] ATMPs—known in the USA as human cells, tissues, and cellular and tissue-based products (HCT/Ps) regulated under Section 351 of the PHS Act and/or the FD&C Act[6]—are subdivided into four categories, of which two are relevant in the context of hematopoietic cellular therapies: somatic cell therapy medicinal products (SCTMP) and gene therapy medicinal products (GTMP). Examples of SCTMP include ex vivo expanded autologous or allogeneic stem cells (de Lima et al. 2012; Delaney et al. 2010), mesenchymal stem cells (Le Blanc et al. 2008), and allogeneic T lymphocytes depleted of alloreactive T cells (Andre-Schmutz et al. 2002). Examples of GTMP include autologous or allogeneic chimeric antigen receptor (CAR)-T cells (Schuster et al. 2017; Neelapu et al. 2017;

[1] https://ec.europa.eu/health/blood_tissues_organs/tissues_en. Accessed 10 July 2023.

[2] www.jacie.org. Accessed 10 July 2023.

[3] https://health.ec.europa.eu/blood-tissues-cells-and-organs/overview/proposal-regulation-substances-human-origin_en. Accessed 10 July 2023.

[4] https://www.europarl.europa.eu/doceo/document/E-9-2022-002133_EN.html. Accessed 10 July 2023.

[5] http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2007:324:0121:0137:en:PDF. Accessed 10 July 2023.

[6] https://www.fda.gov/biologicsbloodvaccines/tissuetissueproducts/regulationoftissues/ucm150485.htm. Accessed 10 July 2023.

Abramson et al. 2020; Berdeja et al. 2021; Munshi et al. 2021; Wang et al. 2020) and autologous CD34+ cells genetically engineered to express a miniglobin gene and designed to treat inherited β-globin disorders (Cavazzana-Calvo et al. 2010; Ribeil et al. 2017; Locatelli et al. 2022) or engineered to re-express fetal hemoglobin as a substitute to β-globin (Frangoul et al. 2021). The regulation was designed in part to foster the competitiveness of European pharmaceutical companies in this emerging field, but the number of ATMPs that have received a centralized marketing authorization has remained relatively low and with poor overall commercial success until the first two autologous CAR-T cells targeting CD19 were approved by EMA in the summer of 2018, approximately 1 year after these two products were approved by the FDA. In the HCT field, it was not until 2015 that an ATMP of interest reached the market with authorization given for Zalmoxis® (allogeneic T cells engineered to express a suicide gene) (Ciceri et al. 2009). Production, distribution, and administration of ATMPs imply a totally different organization than that used for HCT, with manufacturing at a central facility in compliance with good manufacturing practices (GMP) (Wang and Rivière 2016; Wang and Riviere 2017), a version of which was recently released by the European Medicines Agency (EMA) to specifically deal with manufacturing of ATMPs.[7] Since a majority of the ATMPs that progress to authorization or at least to clinical trials are manufactured from autologous mononuclear cells, starting material is currently procured by hospital- or blood bank-operated apheresis facilities creating a peculiar situation in which a product starts under one regulation, Tissues and Cells Directive and likely the SoHO Regulation in the near future, before transitioning to another, ATMP and Pharma Regulations, and where a hospital acts as a service provider to industry, an interaction that requires further definition of the respective responsibilities and liabilities.

[7] https://ec.europa.eu/health/sites/health/files/files/eudralex/vol-4/2017_11_22_guidelines_gmp_for_atmps.pdf. Accessed 10 July 2023.

Publication of Regulation 1394/2007 created a situation in which cell- or tissue-based therapeutic products that were previously prepared and delivered through a POC organization similar to that for cell transplants are now classified as ATMPs. This had a limited impact in the field of hematopoietic cellular therapies, although some cell-based products such as allogeneic T cells with specific anti-CMV activity engineered through the IFN-gamma-catch technology (Feuchtinger et al. 2010) were affected. In recognizing that many potential ATMPs were used for limited numbers of patients and with no commercial motivation, Regulation 1394/2007 created the so-called hospital exemption (HE) under Article 28 exempting from authorization requirements those ATMPs manufactured in hospitals, universities, or start-up companies where the medicine is prescribed for individual patients under the care of a medical practitioner. This manufacture should occur on a non-routine basis according to specific quality standards (GMP) (Vives et al. 2015), and the ATMP should be used in a hospital and only within the same member state. National authorities oversee the approval of HE products which has resulted in significant variations between member states in how it is applied and which has led to criticism from both industry and academia that it is unclear and inconsistent. Recently, Spanish investigators have developed several CAR-T cells that are structurally and functionally comparable to approved commercial products (Ortiz-Maldonado et al. 2021; Oliver-Caldes et al. 2023), one of which, ARI-0001, has been conditionally approved by EMA (Trias et al. 2022). The Pharma Regulation is also undergoing revision at the EU Commission, and the HE is one of the key aspects that is examined in the context of public consultations with all interested stakeholders.

Access to ATMPs including cellular therapies is likely to be a particular challenge for patients, healthcare professionals, and national health systems due to the high costs of the medicinal products but also of apheresis sessions, inpatient and outpatient stays necessary to administer bridging chemotherapy when needed, lymphodepleting or conditioning regimen ahead of the infusion of

SCTMP or GTMP, and preventing or treating the many side effects (Hayden et al. 2022). Even access to decades-old HCT remains strongly associated with higher-income countries (Gratwohl et al. 2015). One potential effect of limited access is the so-called stem cell tourism whereby patients with sufficient financial resource travel to centers outside their own districts or countries in order to get access to these treatments, whether with unproven efficacy or safety or while not being fully informed to make this important decision. The International Society for Cellular Therapy (ISCT) leads the publication of patient advice and other documentation on this phenomenon.[8]

Academic facilities, including stem cell transplant practitioners at large, should strive to remain active players in the development of ATMPs. Academia remains very active in the early phases of clinical trials designed to evaluate innovative SCTMP and GTMP as potential complements, substitutes, or bridges to historical forms of hematopoietic cell transplants (Pearce et al. 2014). Many public institutions have invested significant resources to upgrade their processing facilities to GMP-compliant levels, thus allowing for small-scale POC manufacturing of investigational medicinal products to support phase I and possibly phase II studies, often with the potential that industry will take over in case that promising results warrant further development (de Wilde et al. 2016a). It is important to keep in mind that the development of a medicinal product goes well beyond compliance with GMP during the manufacturing step, but also involves the production of a broad array of preclinical data to support initiation of early clinical trials, and is thus strikingly different from process validation and optimization that are now routinely requested from TEs. Furthermore, academia has to become a proactive stakeholder in the regulatory area by engaging with the authorities, sharing their know-how, and voicing their opinion (de Wilde et al. 2016b). The GoCART Coalition that was recently co-founded by both the EBMT and the European Hematology Association (EHA) in part serves this purpose and aims at facilitating patients' access to CAR-T cells and other cellular immunotherapies through the engagement of multiple stakeholders.[9]

The field is moving at a fast pace and is now well beyond the proof of concept that tissue-based or cell-based medicinal products can be manufactured by a "conventional" pharmaceutical company (Locke et al. 2017), although with continued reliance on critical contributions from academic facilities, e.g., basic science and provision of starting materials. Some of these innovative medicinal products have demonstrated remarkable clinical efficacy for severe or debilitating diseases although sometimes at the expense of equally remarkable toxicity. In 2022, barely 4 years after initial approvals supported with small single-arm international multicenter trials, positive results were published for two out of three large randomized international multicenter trials comparing autologous CAR-T cells targeting CD19 against the standard of care (SOC) including salvage chemotherapy completed with consolidation high-dose chemotherapy supported with autologous hematopoietic cell transplantation in chemo-sensitive patients (Bishop et al. 2022; Kamdar et al. 2022; Locke et al. 2022); superiority of CAR-T cells in the ZUMA-7 (Locke et al. 2022) and TRANSFORM (Kamdar et al. 2022) studies led to the approval of these treatments in second line for patients with refractory or early first relapse diffuse large B-cell lymphoma (DLBCL), heralding the end of a medical practice that arose approximately 30 years earlier (Philip et al. 1995). The rapid adoption of CAR-T cells in real-world practices (Passweg et al. 2020), with the first published reports from continental or national registries demonstrating comparable results with results from the original registration studies (Bachy et al. 2022; Pasquini et al. 2020), supports robust scientific and financial investments in the field, translating in a growing number of investigational cellular immunotherapies being evaluated (Saez-Ibanez et al. 2022). These developments still carry many

[8] http://www.celltherapysociety.org/page/UCT. Accessed 21 Feb 2018.

[9] https://thegocartcoalition.com. Accessed 10 July 2023.

regulatory and operational uncertainties, including the sustainability of multiple and parallel supply chains for different medicinal products, that create an increasing workload for academic facilities and place a growing financial burden on public and private healthcare payers. The availability of multiple types of hematopoietic cellular therapies at the same time when other categories of therapies also reach the market in large numbers contribute to the growing complexity in the evaluation of the medical value of these innovative and expensive therapies, in patients who have received and will receive multiple lines of treatments over many years of care (Chabannon et al. 2015). In this context, organization of long-term follow-up is challenging.[10] Academia through continental registries such as EBMT[11] will continue to play a key role with data and know-how that will be very useful not just for researchers but also for industry, healthcare regulators, and payers.

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[10] http://www.ema.europa.eu/ema/index.jsp?curl=pages%2Fregulation%2Fgeneral%2Fgeneral_content_000658.jsp&mid=WC0b01ac0580961211.

[11] https://www.ebmt.org/registry/car-t-data-collection-initiative. Accessed 10 July 2023.

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Part VIII

Specific Modalities of HCT and Management

Topic Leaders: Anna Sureda, Nicolaus Kröger and Andrew Gennery